

Technology #02

Adaptive Valve Profile for Postural ICP
Regulation

Engineering Technology-Transfer Dossier
CONFIDENTIAL

Technology Maturity: ENGINEERING_DEFINITION
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Not physically validated. Not transfer-ready.

BUYER DECISION PAGE

WHAT IS IT?

Adaptive Valve Profile for Postural ICP Regulation

WHY DOES IT MATTER?

ICP excursions during postural changes cause patient symptoms

WHAT IS ACTUALLY ESTABLISHED?

Engineering definition with 4 governing equations, 11 design inputs, 3 failure modes. Computational model exists. No physical prototype.

WHAT IS NOT ESTABLISHED?

Physical validation, manufacturing process, regulatory pathway, IP ownership.

WHY MIGHT A BUYER CARE?

Opportunity in Fluid Mechanics + Adaptive Valve Design. 5-step build plan with specific test articles and acceptance criteria. Target buyer: Neuroshunt manufacturer with active valve program.

WHAT DOES THE BUYER GET?

Adaptive valve concept + mechanism description; Controller algorithm specification (MODELLED, not production-firmware); Variable-area orifice governing equation + sensitivity analysis; V0 bench prototype BOM (COTS sensor + custom valve body + candidate actuators); This engineering dossier with external precedent + failure analysis + build plan

WHAT DOES THE BUYER HAVE TO BUILD?

Production valve design (dimensioned drawings, tolerances, material selection frozen); Actuator technology selection and validation; Production controller firmware + software V&V; per IEC 62304; Manufacturing process for variable-area valve; Regulatory submission (likely 510(k) with substantial equivalence, or De Novo if no predicate); Clinical validation evidence (IDE trial); Long-term reliability testing (ISO 14708-1 for active implants)

WHAT IS THE NEXT DECISIVE EXPERIMENT?

Variable-area valve prototype (passive, no actuator)

WHAT WOULD MAKE US KILL IT?

Adaptive response time exceeds postural transient duration (cannot respond within seconds)

WHAT TRANSACTION COULD MAKE SENSE?

Sponsored Validation

What Exists Today

Existing:

- + Documented mechanism with governing equations
- + Computational model
- + 5 external evidence sources
- + Design inputs and outputs
- + Failure analysis
- + 5-step engineering build plan
- + Transfer boundary definition

Does Not Yet Exist:

- Physical prototype
- Qualified manufacturing process
- Clinical data
- Verified IP ownership
- Regulatory clearance
- Commercial product

Buyer Can Commission:

- > Specific bench experiment (see build plan)
- > Prototype fabrication
- > Materials study
- > Manufacturing feasibility study
- > Regulatory pre-submission

Development / De-risking Ladder

CURRENT STATE: ENGINEERING_DEFINITION

NEXT ENGINEERING STEP: Variable-area valve prototype (passive, no actuator)

NEXT VALIDATION STEP: Bench prototype validation per build plan

NEXT REGULATORY STEP: Pre-submission to FDA for pathway classification

COMMERCIALIZATION GATE: Clinical validation evidence required

TRANSFER GATE: Transfer-ready when physical validation + regulatory pathway + IP verified

Engineering Investment Ladder

Next \$5K action: WP-01 - Variable-area valve prototype (passive, no actuator) - 4 weeks

Basis: Build plan WP. Scope: Q vs dP characterization. Uncertainty: Effort estimate, not quoted.

Next \$25K action: WP-03 - Closed-loop controller - 6 weeks

Basis: Build plan WP. Scope: Closed-loop tracking error. Uncertainty: Effort estimate, not quoted.

Next \$100K action: WP-05 - Biocompatibility specimens - 12 weeks (external lab)

Basis: Build plan WP. Scope: Cytotoxicity, sensitization, irritation. Uncertainty: Effort estimate, not quoted.

Top 5 Buyer Diligence Questions

Q1: Is the mechanism physically credible for the intended application?

The governing model includes 4 equations. Key assumptions: CSF is Newtonian (valid at low shear rates in shunt flow); Flow is quasi-steady (valve response time > flow timescale); Catheter geometry is rigid (compliance UNKNOWN); No cavitation (validated by Re and pressure range). UNKNOWN: whether assumptions hold in vivo.

Q2: What evidence supports the core claim?

5 external evidence sources. All are external precedent, not physical validation of this specific invention.

Q3: What is the regulatory pathway?

UNKNOWN. Not yet determined. Requires pre-submission to FDA. No predicate identified for some packages.

Q4: Who owns the IP?

UNKNOWN. IP ownership not verified. Requires legal counsel engagement.

Q5: What would it cost to validate?

First experiment: Variable-area valve prototype (passive, no actuator). Estimated effort from build plan. Full validation cost: COST TO BE QUOTED.

Commercial Interest

TARGET APPLICATION: CSF shunt systems for hydrocephalus management

CURRENT ALTERNATIVE: Standard fixed-pressure or programmable CSF shunt valves

MECHANISM DIFFERENCE: Valve whose opening pressure profile adapts to ICP trend via trend-feedback control

POTENTIAL VALUE DRIVER: Addresses: ICP excursions during postural changes cause patient symptoms

ADOPTION BARRIER: Requires physical validation, regulatory clearance, and manufacturing qualification

DEVELOPMENT BARRIER: No physical prototype exists; manufacturing process not established

Licensee Capability Fit

IDEAL BUYER: Neuroshunt manufacturer with active valve program

REQUIRED ENGINEERING CAPABILITY: Catheter design, fluid mechanics / domain-specific engineering

REQUIRED MANUFACTURING CAPABILITY: Medical tubing extrusion, injection molding, cleanroom assembly

REQUIRED REGULATORY CAPABILITY: FDA 510(k) or PMA submission experience

REQUIRED CAPITAL: Validation: \$50-200K. Full development: \$500K-2M (estimates, not quoted)

REQUIRED MARKET ACCESS: Neurosurgery market channels; hospital purchasing relationships

1. Technology Description

Domain: Fluid Mechanics + Adaptive Valve Design

Disciplines: Computational fluid dynamics (CFD), Valve mechanics, Control systems, Polymer actuator engineering

Architecture: CSF shunt valve whose opening pressure profile adapts to ICP trend rather than fixed cracking pressure; reduces postural pressure excursions

Subsystems:

SS-01: Adaptive valve seat - Variable-area flow orifice whose area $A(P, dP/dt)$ depends on pressure and pressure trend

SS-02: Pressure-trend sensor - Measures ICP(t) and computes dP/dt for controller input

SS-03: Actuator (candidate: MEMS electrothermal or shape-memory polymer) - Adjusts valve seat area based on controller signal

SS-04: Controller (P or PI) - Maps $(P, dP/dt)$ -> actuator command to track target opening profile

2. Mechanism Architecture

Physical Changes: Valve orifice area A adapts: when dP/dt indicates postural transient (positive trend with postural change), the valve reduces conductance briefly to attenuate over-drainage; when ICP rises monotonically, conductance increases to maintain drainage.

Key Physics: Variable-area orifice: $Q = C_d * A(dP, dP/dt) * \sqrt{2 dP / \rho}$. The 'adaptive' component is the $A(dP, dP/dt)$ dependence — a non-linear feedback on the orifice equation.

3. Governing Engineering Model

Summary: Variable-area hydraulic orifice with trend-feedback control. The valve is modeled as a non-linear orifice whose area A is a function of measured pressure and its time-derivative.

Equations:

$Q = C_d * A * \sqrt{2 * dP / \rho}$ [turbulent/orifice flow]

$Q = (\pi * r^4 * dP) / (8 * \eta * L)$ [laminar flow through circular conduit]

$Re = (\rho * v * D) / \eta$ [laminar if $Re < 2300$]

$A_{\text{actuator}}(P, dP/dt) = A_0 + k_p * (P - P_0) + k_d * dP/dt$ [proposed controller law, MODELLED]

Assumptions:

- CSF is Newtonian (valid at low shear rates in shunt flow)
- Flow is quasi-steady (valve response time > flow timescale)
- Catheter geometry is rigid (compliance UNKNOWN)
- No cavitation (validated by Re and pressure range)

Boundary Conditions:

- Inlet: ICP in [5, 40] mmHg
- Outlet: intra-abdominal pressure ~5-15 mmHg supine, lower upright
- Postural head differential: 0-50 cm H₂O (~0-37 mmHg)

Failure Regimes:

- Actuator saturation (max area reached) -> reverts to fixed-valve behavior
- Filter lag > postural transient duration -> adaptation ineffective
- Sensor noise > dP/dt signal -> controller unstable

4. Design Inputs

ID	Input	Value	Origin
DI-001	Clinical need	ICP excursions during postural changes cause patient symptoms and possible complications (R332 problem statement)	VERIFIED
DI-002	Functional requirement	Valve opening profile adapts to ICP trends to reduce postural pressure excursions	MODELLED
DI-003	Flow regime	Laminar, $Re \ll 2300$ at typical CSF flow rates	COMPUTATIONALLY_SUPPORTED
DI-004	Pressure range	ICP 5-20 mmHg normal, postural transients up to 40 mmHg	VERIFIED
DI-005	ICP safety threshold	$P_{ICP} \leq 20$ mmHg (INV-1, MODEL_DERIVED)	MODELLED
DI-006	Minimum drainage floor	$Q \geq 0.05$ mL/min (INV-2, MODEL_DERIVED)	MODELLED
DI-007	Biocompatibility (ISO 10993)	UNKNOWN	UNKNOWN
DI-008	Sterilization compatibility	UNKNOWN	UNKNOWN
DI-009	EMC	UNKNOWN	UNKNOWN
DI-010	Response time	Target < 5 s (postural transient timescale)	MODELLED
DI-011	Controller type	PI controller with trend feedback (proposed)	PROPOSED

5. Design Outputs

ID	Description	Status	Missing Inputs
DO-001	Valve seat geometry (variable-area orifice)	ABSENT	A_{max} ; A_{min} ; valve seat profile; actuator mechanism
DO-002	Pressure sensor specification	CONCEPTUAL	
DO-003	Controller firmware	MODELLED	
DO-004	Actuator specification	ABSENT	Actuator technology choice; force/displacement requirement; power bud

6. Critical Design Parameters

Name	Value	Unit	Basis	Verification
Cracking pressure P_crack	UNKNOWN	mmHg	Design choice	Bench measurement per ISO 7197
Maximum valve area A_max	UNKNOWN	mm^2	Design choice	Geometric inspection + flow bench
Response time tau_valve	UNKNOWN	s	Actuator-dependent	Step response test
Controller gains k_p, k_d	UNKNOWN	dimensionless, min/s	Tuned via simulation	Closed-loop stability test

7. Failure Modes

Mode	Mechanism	Mitigation	Verification	Residual Uncertainty
Actuator jam	Mechanical obstruction or electrical failure	Fail-safe pressure mode (fallback)	Defeat plug, verify fallback	UNKNOWN whether fallback meets clinical need
Sensor drift	Long-term pressure sensor drift	Periodic recalibration protocol (per UNKNOWN specifics)	Algorithmic drift UNKNOWN	Drift magnitude in CSF environment UNKNOWN
Postural misclassification	Controller misclassifies postural transition	Multi-sensor fusion (postural response)	Behavioral simulation	Classifier accuracy UNKNOWN

8. Failure Analysis

Failure Mode	Mechanism	Mitigation	Residual Uncertainty
Actuator jam	Mechanical obstruction or electrical failure	Fail-safe fixed-pressure fallback	UNKNOWN fallback efficacy
Pressure sensor drift	Long-term offset drift	Recalibration protocol (UNKNOWN specifics)	Drifts magnitude in CSF UNKNOWN
Battery depletion (if active)	Energy source exhausted	Low-power design + indicator	UNKNOWN battery life given duty cycle
Postural misclassification	dP/dt filter misclassifies transient	Multi-sensor fusion (postural + pressure)	Classifier accuracy UNKNOWN
Mechanical fatigue of valve seat	Cyclic actuation causes material fatigue	Material selection + lifetime derating	Cycle count to failure UNKNOWN

9. Verification Strategy

ID	Requirement	Method	Acceptance	Result
V-001	Valve tracks target opening profile	Bench flow loop	Steady-state error < 2 mmHg	NOT_TESTED
V-002	Response time < 5 s to postural step	Step input test	10-90% rise < 5 s	NOT_TESTED
V-003	Minimum drainage floor maintained	Worst-case flow measurement	Q >= 0.05 mL/min	NOT_TESTED

10. Validation Strategy

ID	Requirement	Method	Acceptance	Result
VAL-001	Reduced ICP excursion vs fixed valve in patient trial (IDE required)	Clinical trial (IDE required)	Statistically significant reduction	NOT_PERFORMED

11. Materials

Component	Candidate Material	Source	Verification Required	Status
Valve body	Silicone elastomer	Standard CSF shunt material (K052009901ethkephthalate)	ISO 10993 + fatigue + activation	CANDIDATE
Valve seat (variable-area shape-memory)	Shape-memory polymer (SMP)	Published SMP valve literature	ISO 10993 + fatigue + activation	CANDIDATE - target selected
Pressure sensor diaphragm	Silicon (MEMS)	Standard MEMS pressure sensor	ISO 10993 + drift characterization	CANDIDATE

12. Bill of Materials

Item	Description	Qty	Type	Material	Supplier	Criticality	Verification
01	Valve body housing	1	CUSTOM_COMPONENT	UNKNOWN — likely silicone/aluminum	UNKNOWN	CRITICAL (extensive precedent)	Biocompatibility
02	Pressure sensor	1	COTS_CANDIDATE	MEMS (COTS)	UNKNOWN — to be evaluated	HIGH	Calibration + drift test
03	Actuator (MEMS or shape-memory)	1	CUSTOM_COMPONENT	UNKNOWN — depends on technology	UNKNOWN	CRITICAL	Lifetime + force calibration
04	Microcontroller	1	COTS_CANDIDATE	MCU (COTS)	Multiple (TI, STM, Microchip)	MEDIUM	Software validation + EMC

13. Manufacturing

Status: ENGINEERING_CANDIDATE — no process validated

Candidate Processes:

- Silicone injection molding: Standard CSF valve manufacturing - Variable-area mechanism may require modified process
- MEMS fabrication (if actuator is MEMS): Standard MEMS foundry process - Requires foundry partnership
- Laser welding (housing assembly): Standard implantable device assembly - Hermetic seal required

14. External Evidence

Source 1: Relationship between valve opening pressure, body

URL: <https://pubmed.ncbi.nlm.nih.gov/15458593>

Excerpt: by M Bergsneider · 2004 · Cited by 77 — This study indicates that very high valve opening pressure settings may be optimal for the initial treatment of normal pressure hydrocephalus .

Source 2: Paradoxical response of intracranial pressure to shunt valve

URL: <https://link.springer.com/article/10.1007/s00701-020-04462-y>

Excerpt: by L D'Antona · 2020 · Cited by 3 — This study describes the effect of adjusting the shunt valve setting on ICP in a cohort of patients with complex symptom management.

Source 3: Understanding Shunt Systems

URL: <https://www.hydroassoc.org/shunt-systems>

Excerpt: Adjustable (programmable) valves regulate the ICP based on a pressure setting, like the fixed pressure valve, but the setting can be adjusted by your doctor

Source 4: Impact of shunt adjustments and body positions

URL: https://www.researchgate.net/publication/263394882_Intracranial_pressure_in_hydrocephalus_Impact_of_shunt_adjustments_and_body_positions

Excerpt: Noninvasive manipulation of programmable shunt valves may allow customization of intracranial pressure (ICP) dynamics in individual shunted patients. Read more

Source 5: Intracranial Pressure Sensor and Valve to Control

URL: <https://www.semanticscholar.org/paper/fcd155029dbc7fa5a82915822f2289055da96164>

Excerpt: This work is working to develop an implantable pressure sensor that will open a valve to relieve the intracranial pressure (ICP)

15. Transfer Boundary

YOU RECEIVE:

- + Adaptive valve concept + mechanism description
- + Controller algorithm specification (MODELLED, not production-firmware)
- + Variable-area orifice governing equation + sensitivity analysis
- + V0 bench prototype BOM (COTS sensor + custom valve body + candidate actuators)
- + This engineering dossier with external precedent + failure analysis + build plan

YOU MUST DEVELOP:

- > Production valve design (dimensioned drawings, tolerances, material selection frozen)
- > Actuator technology selection and validation
- > Production controller firmware + software V&V; per IEC 62304
- > Manufacturing process for variable-area valve
- > Regulatory submission (likely 510(k) with substantial equivalence, or De Novo if no predicate)
- > Clinical validation evidence (IDE trial)
- > Long-term reliability testing (ISO 14708-1 for active implants)

YOU MUST VERIFY:

- ? Biocompatibility (ISO 10993)
- ? Sterilization compatibility
- ? Mechanical integrity
- ? Regulatory pathway
- ? IP ownership and freedom to operate
- ? Manufacturing process capability

16. Open Questions

- Actuator technology choice (MEMS electrothermal vs shape-memory polymer vs electromagnetic) — UNKNOWN
- Long-term drift of pressure sensor in CSF environment — UNKNOWN
- Controller gains in vivo — UNKNOWN until clinical data
- Biocompatibility of actuator materials in CSF — UNKNOWN (ISO 10993 testing required)
- Battery life if active (or energy harvesting if passive) — UNKNOWN
- Regulatory pathway (Class II with special controls vs Class III PMA) — UNKNOWN until pre-submission to FDA

17. Decisive Next Experiment

WP	Test Article	Equipment	Measurement	Acceptance	Deliverable	Effort
WP-01	Variable-area valve prototype (passive, 1mm ² orifice)	Transducer (0-100 mmHg), pressure sensor, 100 Hz	Flow rate (ml/min) vs pressure (mmHg)	Flow rate < 100 ml/min at 100 mmHg	Passive flow characteristics report	1 week
WP-02	Actuator-integrated valve bench flow loop + actuator	Flow loop, 100 ml/min, 100 Hz	Step response of A(t)	10-90% rise < 5 s	Active valve step-response report	1 week
WP-03	Closed-loop controller	Real-time controller (e.g., dSPACE)	Flow rate vs pressure (mmHg)	Steady-state error < 20%	Closed-loop performance report	1 week
WP-04	Postural simulator	Programmable pressure sensor	Pressure (mmHg) vs posture	Pressure < 100 mmHg	Adapted valve (MODEL_DERIVED target)	1 week
WP-05	Biocompatibility specimen	ISO 10993 test lab	Cytotoxicity, sensitization	ISO 10993 pass	ISO 10993 report	12 weeks (external lab)

18. Buyer Decision Framework

WHY INVEST: Engineering definition exists with 4 governing equations and a 5-step build plan. The next experiment is clearly specified. Target buyer: Neuroshunt manufacturer with active valve program.

WHY WAIT: No physical prototype exists. Regulatory pathway is unresolved. IP ownership is unverified. Manufacturing process is not qualified.

WHY REJECT: Adaptive response time exceeds postural transient duration (cannot respond within seconds)

WHAT WOULD CHANGE THE DECISION: A successful bench experiment demonstrating the core mechanism. A qualified regulatory pathway. A confirmed IP position.

19. Kill Condition

KILL_IF: Adaptive response time exceeds postural transient duration (cannot respond within seconds)

DISCLOSURE

This dossier distinguishes established evidence, external precedent, computational/modelled results, engineering proposals, and unresolved questions. The absence of physical validation should not be interpreted as evidence that the technology will fail; it means the relevant proposition has not yet been established experimentally.

This dossier does not constitute legal, patentability, FTO, regulatory, or investment advice.